

Sensing

Electronic systems allow security and monitoring systems to respond in changing situations, conditions and contexts. Sensors may signal door and window motions, smoke or carbon-monoxide presence, physical disturbances, pressure changes and more. These connected sensors offer a wide range of uses in smart home security and automation applications. Whether through specific sensor components and related op-amps employed to enable battery-powered wireless devices or through ESD protection for high-speed signal lines, engineers require a range of products to safeguard their designs.

Now, let us examine how sensor components and related solutions lead to excellent security monitoring design.

Sensor devices provide real-time system protection, feedback control and high accuracy system monitoring. Position sensors enable engineers to determine absolute and relative positions, including angle, presence, proximity, distance, flow, level and velocity. Such sensors as light, image and analogue front-end (AFE) help designers capture a broad range of wavelengths.

Because your human eyes cannot be everywhere, it is helpful to have another set of eyes on your home to establish extra protection. One option a designer might consider employing is an ambient light sensor. An ambient light sensor measures the intensity of visible light. Spectral response of the sensor tightly matches the photopic response of the human eye and includes significant infrared (IR) rejection (Fig. 2).

Still, looking for home intruders is not the only required sensing function in many security monitoring systems. Often a change in the ambient environment warrants monitoring. A change in humidity may indicate that an environmental shift is in process that may cause significant damage to valuables within a monitored location. In such a situation, a humidity sensor may do the job. It can provide excellent measurement accuracy at very low power levels and may also include an integrated temperature sensor.

In isolation, sensors alone may not be able to complete the job. Often, these need a signal boost using amplifiers so that downstream electronics can further manipulate and evaluate their electronic information. Ultra-low-power, operational amplifiers find employment in sensing applications with battery-powered, wireless and low-power wired equipment. Such ultra-low-power, operational amplifiers help reduce power consumption in equipment like carbon-monoxide detectors, smoke detectors and passive infrared (PIR) motion detectors where operational battery-life is critical.

Command and control

At the core of any home automation security and monitoring application, there is a need to have command and control over electronic signals. A proper command and control system allows for the achievement of greater energy and operational efficiency in the system. By incorporating products and technology with the brains to better monitor and regulate systems, including lighting, windows and doors, video cameras and more, a home becomes more safe and secure for its occupants.

An effective way to utilise the command and control function is to use a controller, specifically a microcontroller unit (MCU). MCUs are useful in embedded devices, as these contain the processor, memory and peripherals in one integrated unit. Many of today's MCUs may excel at ultra-low-power sensing and measuring. These allow designers to use a command and control platform from which to coordinate their design requirements. Since they possess internal memory, both random access memory (RAM) and read-only memory (ROM), MCUs are well-positioned to deliver fast, flexible and reliable command and control functions without requiring additional off-chip capabilities. Onboard memory and peripherals help MCUs minimise their total power consumption.

Communications

Home automation security and monitoring applications, of which the

increasing need for wireless connectivity to accommodate the IoT is driving, require a new dimension of wireless connectivity security. Upon proper authorisation, only where appropriate, data must undergo proper protection by allowing proper send and receive transmissions.

In home automation security and monitoring designs, wireless MCUs, which fully support Bluetooth, Zigbee, low-power wireless personal-area networks (LoWPAN), Thread, Wi-Fi, sub-1GHz, Sigfox or other protocols must by design provide highly-secure, cloud-ready solutions for IoT accessibility. Additionally, wireless MCUs require an input voltage filtering/decoupling network and an output impedance matching network for the radio frequency (RF) antenna.

Today, wireless MCUs can offer designers access to distinct wireless protocol modes. These may offer a blend of long/robust range, low power and high data rates. By design, these often consume very low amounts of power, allowing for battery operation. These may come with sensors that help them recognise when these should be active and communicating, or when these should be in sleep or wait state.

Wireless MCUs give engineers greater flexibility, as the designers may consolidate the command and control function with communication function ultimately to offer high levels of integration. Such levels of integration will simplify designs and ensure continuously reliable communication.

Conclusion

Security and monitoring in the field of home automation continue to evolve. New electronic designs and solutions continue to accelerate this trend. Now that many homes are automated, smart and connected, electronic designers need to continue to offer improvements on issues related to power, sensing, command and control, and communications. A design engineer's imagination is never automated. Consider how your imagination may take hold of the future, and continue to make your future and your homes more convenient, secure and safe. **EFY**